PD-KLP-USA-1

PRINTDECK

INSTRUCTION BOOKLET

For the Printdeck touch screen and Klipper printers



Printdeck is free software: AGPL-3.0, with the Go port of Klipper under GPL-3.0.
The deck draws every picture of this booklet.

https://github.com/beetlebugorg/printdeck

Contents

SET UP

Connect to Wi-Fi 3
Choose your printer 5
Install the Klipper firmware 7

MEET YOUR PRINTDECK

The deck 8
The display of the printer 10
The deck in a browser 11

PRINT A PART

Send a file from the slicer 12
Start a print 13
Watch the print 15
Look around during a print 22

CHANGE THE FILAMENT

Change the filament 23

MOVE AND HEAT

Move the head and heat 29

LEVEL AND CALIBRATE

Level and calibrate 31

LOOK AFTER THE PRINTER

Look after the printer 36
Update the Printdeck 41

WHEN SOMETHING GOES WRONG

When something goes wrong 42
Settings 43

Settings

The file printdeck.txt on the PRINTDECK partition holds the settings. Put the disk in a computer to change it. The deck reads it at each start, then clears the Wi-Fi password in it.

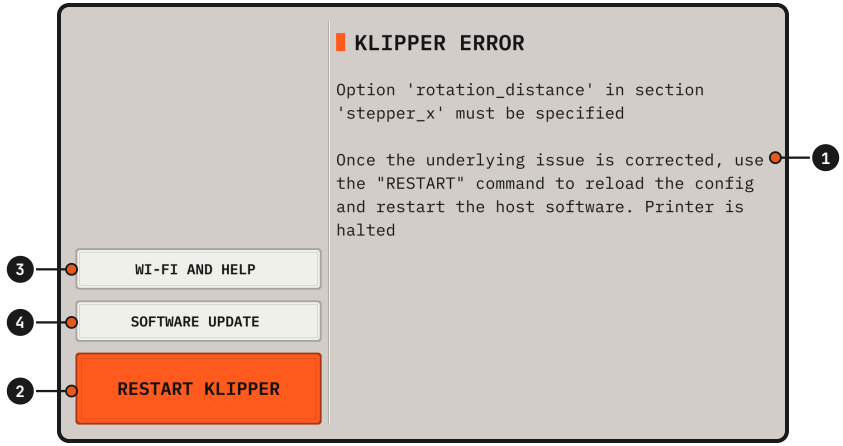
`wifi_ssid, wifi_password, wifi_country`	The Wi-Fi network
`hostname, timezone`	The name on the network, and the time zone
`display_rotate`	90 or 270 for a display that is mounted tall, 0 or 180 for one that is mounted wide
`touch_invert_x, touch_invert_y, touch_swap_xy`	1 turns an axis of the touch screen
`preheat_tool, preheat_bed`	The temperatures of Preheat with no filament set
`night_from, night_to`	The hours of the night colors
`printer.cfg (a file)`	A Klipper config. The deck takes it at the next start.

When something goes wrong

When Klipper stops with an error, the deck shows all the lines of its message.

1. Read the message. It often names the setting in printer.cfg.
2. After a stop or a power cut of the printer, tap Firmware restart.
3. After a change to printer.cfg, tap Restart Klipper.

A Klipper error



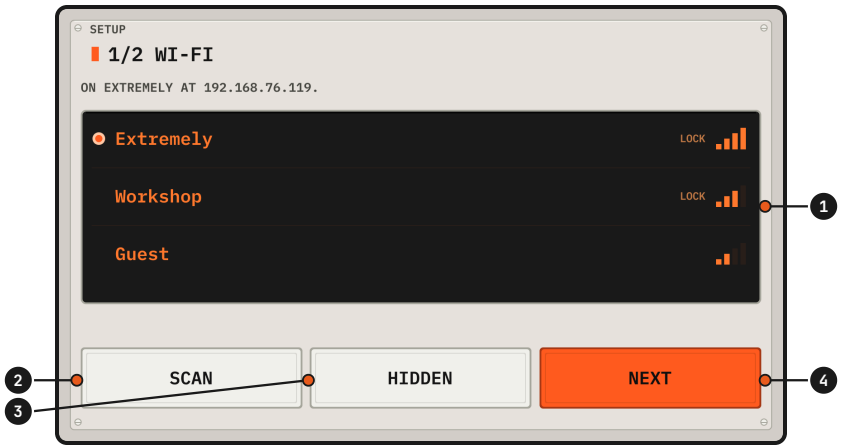
1. **Message.** What Klipper reports, with all its lines.
2. **Restart Klipper.** Restarts Klipper. It reads printer.cfg again.
3. **Wi-Fi and help.** The network, the address and the password of the deck.
4. **Software update.** Installs a new version of Printdeck OS. A new version can fix the problem.

Connect to Wi-Fi

At the first start, the deck makes its password and opens the setup.

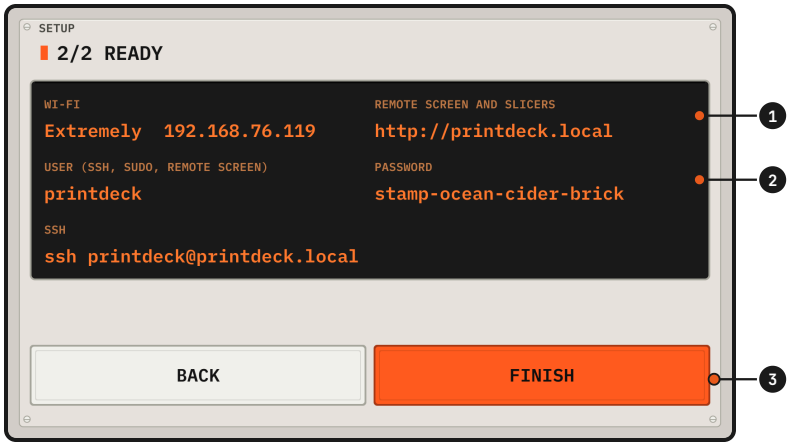
1. Switch on the Printdeck. The setup opens.
2. Tap your Wi-Fi network, then type its password.
3. Wait until the deck is connected, then tap Next.
4. Write down the address and the password, then tap Finish.

The setup: step 1



1. **Networks.** The Wi-Fi networks near the deck, with the signal. A lock shows a network with a password. Tap a network, then type its password.
2. **Scan.** Looks for the networks again. The list also scans by itself.
3. **Hidden.** For a network that does not show its name: type the name and the password.
4. **Next.** Goes to the summary when the deck is connected.

The setup: step 2



1. **Address.** The address of the deck on your network. Open it in a browser for the web pages.
2. **Password.** The password of the web pages. To see it again, open Machine > Wi-Fi and help.
3. **Finish.** Closes the setup. The deck does not open it again.

> You can also set the Wi-Fi before the first start: put the disk in a computer, and fill in printdeck.txt on the PRINTDECK partition.

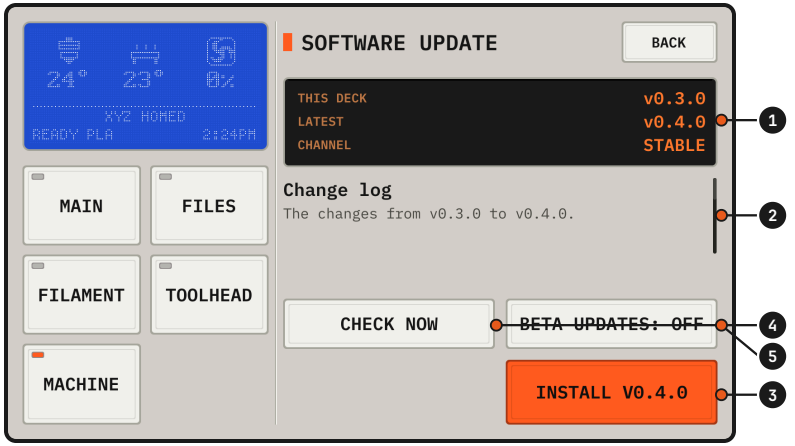
Update the Printdeck

The deck checks for a new version once a day. Machine then shows Software update in orange.

Each version has a signature. The deck installs only a version with a valid signature of Printdeck. It writes the other system slot and restarts into it. If the new system does not start, the deck goes back to the old one.

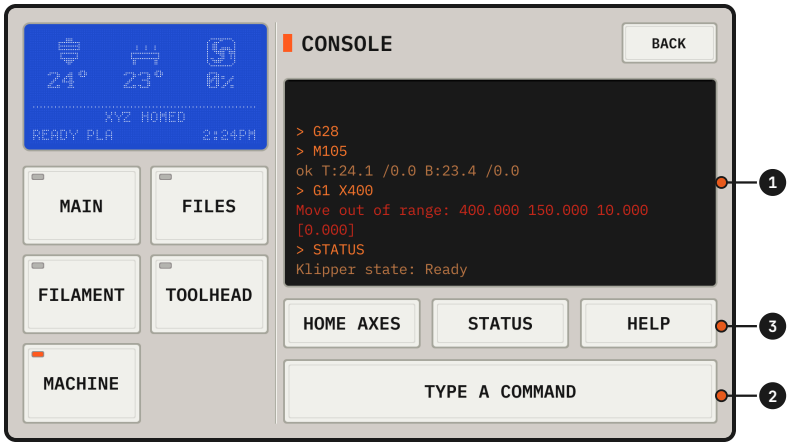
1. Open Machine, then Software update.
2. Read the notes of the new version, then tap Install. The deck does not install during a print.
3. Wait while the deck downloads and writes the update. It restarts into the new version.

Software update



1. **Versions.** The version of this deck, the newest version, and the channel.
2. **Notes.** What the new version changes.
3. **Install.** Downloads the new version, checks it, and installs it, after a question. The deck restarts.
4. **Check now.** Checks for a new version now.
5. **Beta updates.** On: the deck also gets the betas, the versions with new features for tests. Off: only the stable versions.

The console



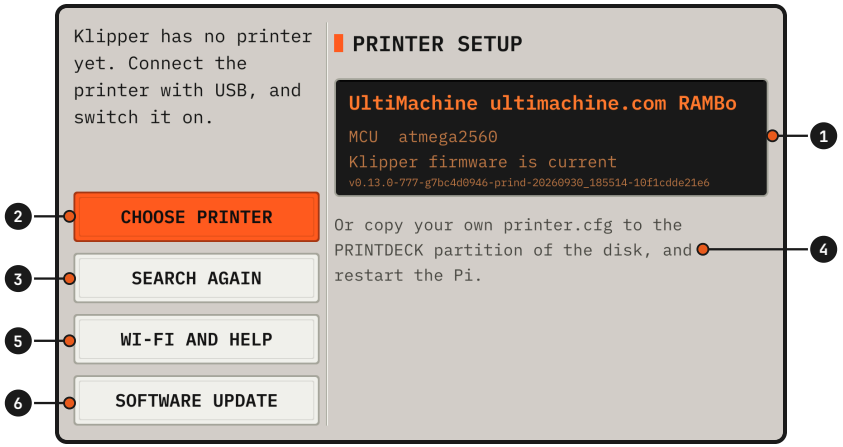
1. **Replies.** Your commands (amber, after >), the replies of Klipper, and errors (red). Drag down to read older lines.
2. **Type a command.** Opens the keyboard. It shows your last command, so you can change it and send it again.
3. **Quick keys.** Home (G28), Status (STATUS) and Help (HELP, the list of the commands).

Choose your printer

A new Printdeck has no printer config. The deck then shows the printer setup.

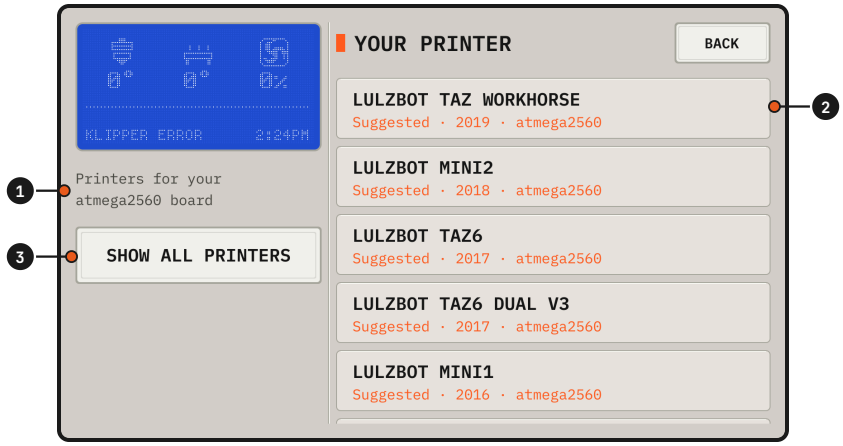
1. Connect the printer to the Printdeck with a USB cable, and switch the printer on.
2. Wait until the deck finds the board. Tap Search again if it does not show.
3. Tap Choose printer. The printers of the brand of your board are at the top.
4. Tap your printer, then Use. Klipper restarts with its config.

No printer yet



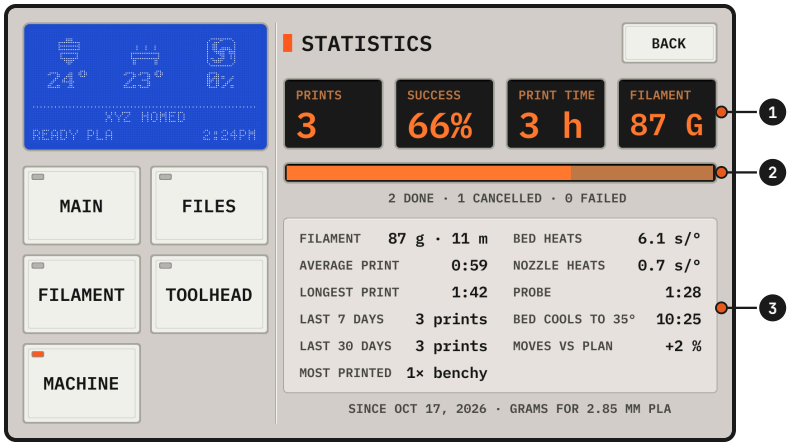
1. **Board.** The board that the deck found on USB, its MCU, and its Klipper firmware.
2. **Choose printer.** Opens the list of Klipper's example configs. See below.
3. **Search again.** Looks for the board again, for example after you connect the printer.
4. **Your own config.** You can also copy your own printer.cfg to the PRINTDECK partition, and start the deck again.
5. **Wi-Fi and help.** The network, the address and the password of the deck.
6. **Software update.** Installs a new version of Printdeck OS. A new version can fix the problem.

Choose your printer



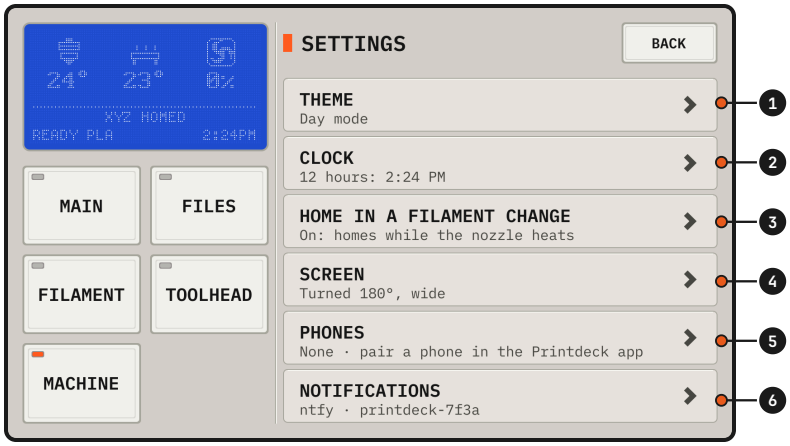
1. **Chip.** The list shows the printers with the same chip as your board.
2. **Printer.** The brand, the model, the year and the chip. Tap your printer. The deck writes its config as printer.cfg, with the port of your board, and restarts Klipper. The old printer.cfg stays as printer.cfg.bak.
3. **Show all printers.** Shows the printers for all chips.

Statistics



1. **Totals.** The number of prints, the success rate, the print time and the filament.
2. **Results.** Done (amber), cancelled and failed (red).
3. **Details.** The filament in grams (for 1.75 mm PLA), the average and the longest print, the prints of the last 7 and 30 days, and the file that you print most.

The Settings page



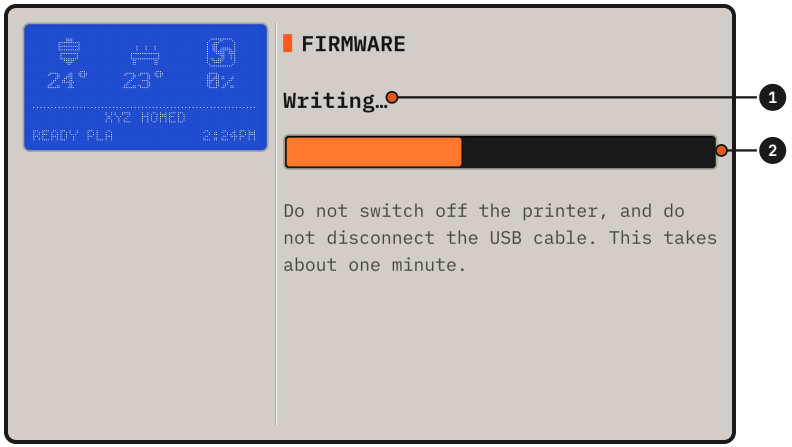
1. **Theme.** The colors of the screen: day, night, or auto (night from 20:00 to 7:00). Tap it to change them.
2. **Clock.** The clock of the screens and the display of the printer: 12 hours (2:24 PM) or 24 hours (14:24). Tap it to change it.
3. **Home in a filament change.** On: the printer homes all axes while the nozzle heats for a filament change. A change during a paused print does not home. Tap it to turn it on or off.
4. **Screen.** The turn of the screen: 0° or 180° for a display that is mounted wide, 90° or 270° for a display that is mounted tall. A tap asks to turn it a quarter turn more, then the deck restarts in the new turn. The touch turns with the screen. A print goes on during the restart. This setting wins over display_rotate in printdeck.txt.
5. **Phones.** The phones with the Printdeck app that get the push messages of the prints. The app pairs with the deck over the network, with the address and the API key. The messages are encrypted from the deck to the phone. Tap the row for the list; a tap on a phone asks to remove it.
6. **Notifications.** Push messages to your phone and watch: print done, print failed, a pause, the end of the first layer, and a roll that runs short. Install the ntfy app on the phone, and subscribe to a topic with a name that is hard to guess. Type the same topic here, or the address of a topic on your own ntfy server. A test message goes out when you save it. An empty topic turns the messages off.

Install the Klipper firmware

The board of the printer must run the Klipper firmware of this Printdeck. The deck writes it through the USB cable.

1. Open Machine, then Maintenance, then tap Firmware. A new board shows Install Klipper firmware on the setup screen.
2. Tap Write. Do not switch off the printer until the deck shows the result.
3. Tap OK. Klipper restarts and connects to the board.

The firmware flash



1. **Step.** Starting the bootloader, Writing, then Checking.
2. **Progress.** The part of the firmware that is done.

> The deck stops Klipper, resets the board into its bootloader, writes the firmware, and reads it back to check it. Then Klipper starts again. The setup screen and the Klipper error screen also show this key when the board has no Klipper firmware or old firmware. Printdeck can flash boards with an atmega2560 and an STK500v2 bootloader now, for example the RAMBo and RAMPS boards.

The deck

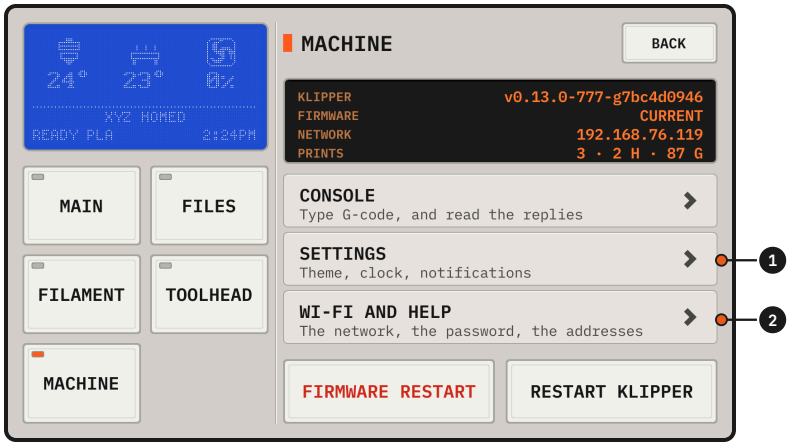
Printdeck is a touch screen for a 3D printer that runs Klipper. It shows the print, and it controls the printer.

When the printer does not print, the deck shows the idle screen. The largest key is the action that you need most: Print a file.

The keys are the keys of a machine. A key with an LED has two states, and its LED is lit while it is on: the page that is open, the material in the printer, Preheat. A key with no LED does an action: an orange ring shows that it took the tap.

Each screen has the same places. At the left, the display of the printer, and under it the key panel: Main (the idle screen), Files, Filament, Toolhead and Machine. At the right, the page that is open: its title and Back at the top, and its actions in one row at the bottom, with the main action at the right.

The Machine page, further down



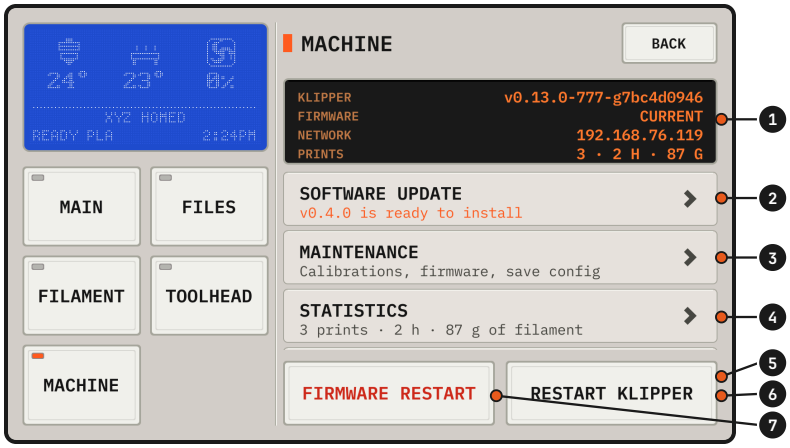
1 **Settings.** The theme, the clock, and the homing in a filament change. See below.

2 **Wi-Fi and help.** The network, the password and the addresses.

Look after the printer

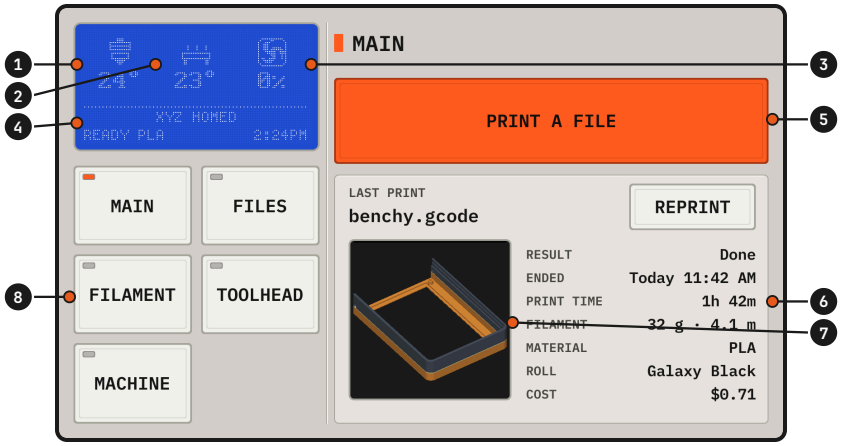
Machine has the information of the printer, the history of the prints, the macros, the console and the settings.

The Machine page



1. **Information.** The version of Klipper, the state of the firmware of the board, the network address, and the totals.
2. **Software update.** The version of the deck, and a new version when there is one. See Update the Printdeck.
3. **Maintenance.** The calibrations, the firmware, and Save config. See Maintenance.
4. **Statistics.** The totals of all prints.
5. **Print history.** Each print, with its result. Tap a print to see its model, its time and its filament, and to print it again.
6. **Restart Klipper.** Restarts the Klipper host, which reads printer.cfg again.
7. **Firmware restart.** Restarts the firmware of the board, for example after a stop.

The idle screen



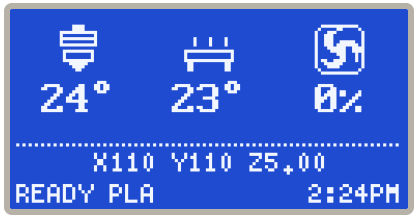
1. **Nozzle.** The nozzle now, and its target under it while the heater is on. The icon is filled while the heater is on, and waves rise while it heats. Tap it to set the temperature.
2. **Bed.** The same for the bed.
3. **Fan.** The part cooling fan. Its blades turn at its speed.
4. **Status.** The position of the head, the state of the printer with the loaded filament, and the clock. This is the status screen of the display of the printer: the same picture on both screens.
5. **Print a file.** Opens the files. See Files.
6. **Last print.** The last print, as on its page of the history: the result, the time, the filament, the roll and the cost. Reprint starts it again, after a question. Preheat and Cool down are in Filament.
7. **Model.** The model of the last print, in the color of its roll. Tap it to open the page of the print in the history.
8. **Pages.** The key panel: Main (this screen), Files, Filament, Toolhead and Machine. The LED of the page that is open is lit. Each page has its own section.

The display of the printer

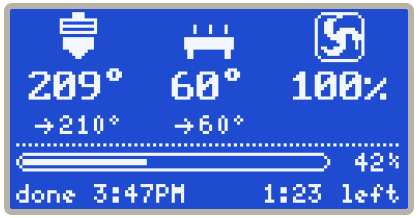
When the printer has a display, the deck draws it. The display shows the status screen, the same picture as the idle screen of the deck.

Turn or click the knob to open the menu. The menu has the same pages and keys as the touch screen: the knob moves the cursor, and a click taps the key. The touch screen shows the cursor with an orange ring. The top item goes back: Status on the home menu, Back on a page. A long click also goes back. With no turn for 20 seconds, the display shows the status screen again.

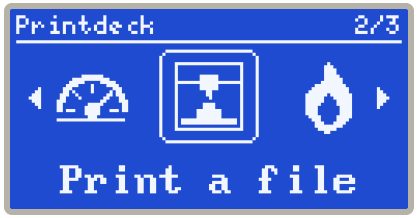
While an action runs, the display shows an animation.



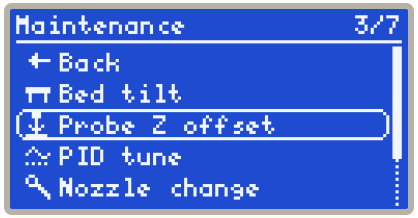
THE STATUS SCREEN



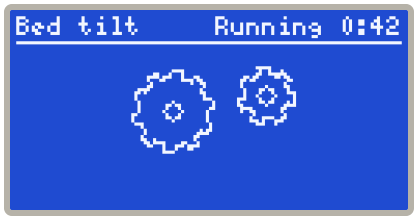
DURING A PRINT



THE HOME MENU: TURN THE KNOB TO CHOOSE

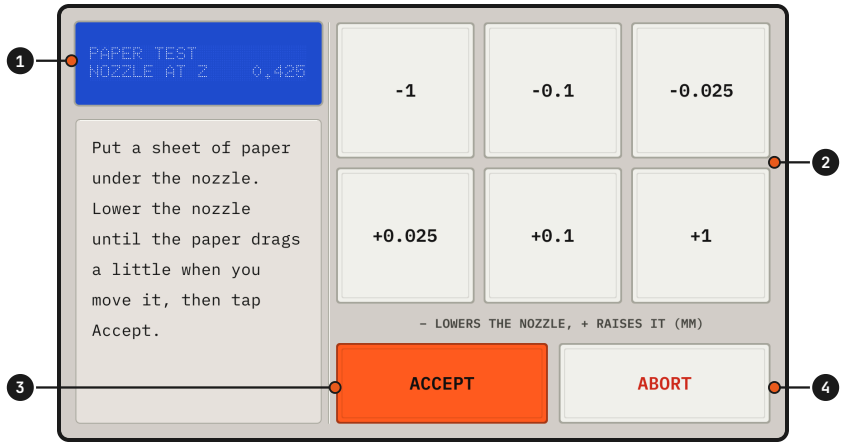


A PAGE: MAINTENANCE



AN ACTION RUNS

The paper test



1. **Height.** The height of the nozzle now.
2. **Steps.** Moves the nozzle down (–) or up (+), in mm.
3. **Accept.** Takes the height. Save config then writes it.
4. **Abort.** Stops the test with no change.

> This screen shows by itself while Klipper runs a paper test, also when it started from a macro.

An action runs



1 **Action.** The time since the start, a bar that moves while the action runs, and the temperatures.

2 **Stop the printer.** Stops the printer at once, after one question. Then use Firmware restart.

While an action runs, the deck shows it on the deck and in each browser, and the other keys are off. The display of the printer shows the name of the action and a moving bar. When the action ends, a short message says that it is done. When it fails, the deck shows the error of Klipper until you tap OK.

The deck in a browser

Open http://printdeck.local in a browser on the same network, and log in with the password of the deck. The browser shows the same screens as the touch screen, sharp at the pixel density of the phone. Each browser has its own screens, so a phone can show the files while the deck shows the print.

On a phone, use Add to Home Screen: the deck then opens as an app, with no browser bars.

The page http://printdeck.local/system shows the version, the services, the logs and the API key for slicers. It also installs an update: choose the update file, then Install. The deck writes the other system slot and restarts into it. If the new system does not start, the deck goes back to the old one.

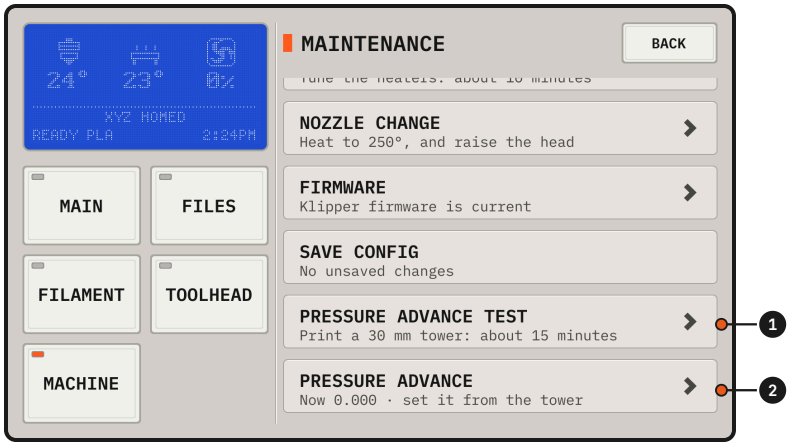
Send a file from the slicer

The deck answers the OctoPrint API that slicers use to send files and to start prints.

In Cura, add the deck with the OctoPrint Connection plugin. The deck asks "Allow Cura to use the Printdeck?". Tap Allow.

In OrcaSlicer or PrusaSlicer, add a printer of the type OctoPrint with the address http://printdeck.local, and the API key of the system page.

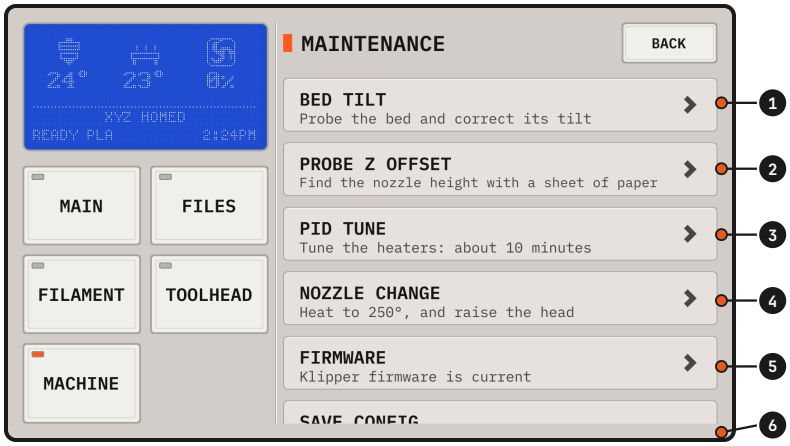
The Maintenance page, further down



1. **Pressure advance test.** Prints a square tower, 30 mm high, with thin walls. The pressure advance goes up with the height: 0.005 for each mm. Look at the corners: low on the tower they bulge, high on the tower they are rounded. Measure the height where they are sharpest.

2. **Pressure advance.** Type the height of the sharpest corners. The deck sets the pressure advance from it, and writes it in printer.cfg.

The Maintenance page



1. **Bed tilt.** Probes the bed, and corrects its tilt (BED_TILT_CALIBRATE). A printer with a bed mesh shows Bed mesh.
2. **Probe Z offset.** The paper test for the height of the nozzle (PROBE_CALIBRATE).
3. **PID tune.** Heats the nozzle to 210° and the bed to 60°, and finds the PID settings of each heater (PID_CALIBRATE). Save config then writes them.
4. **Nozzle change.** Heats the nozzle to 250° and raises the head, to change the nozzle.
5. **Firmware.** Writes the Klipper firmware of this Printdeck to the board of the printer. Orange text shows old firmware. See below.
6. **Save config.** Writes the results of the calibrations into printer.cfg. Klipper restarts. Orange text shows unsaved results.

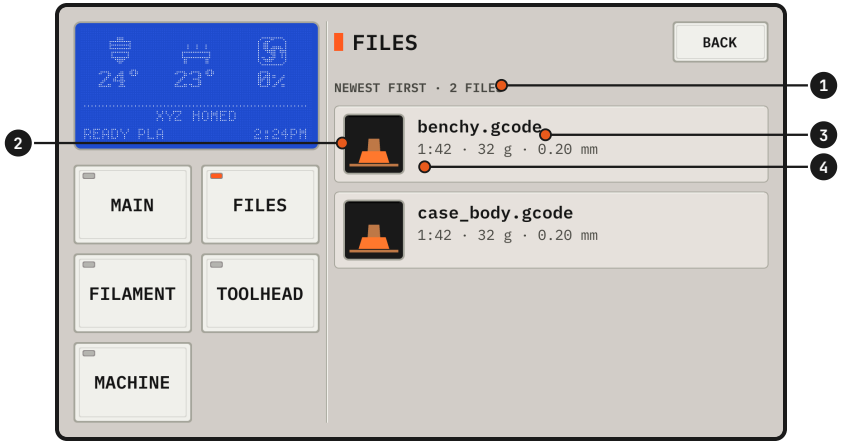
Start a print

The files are the G-code files that the slicer sent, or that you uploaded in the browser. The newest file is at the top.

Each file shows what the slicer wrote into it: the print time, the filament and the layer height. Cura, PrusaSlicer, OrcaSlicer and Bambu Studio write this data. A thumbnail shows when the slicer adds one.

1. Tap Print a file on the idle screen.
2. Tap the file. Check the print time, the filament and the temperatures.
3. Tap Print, then Print again.

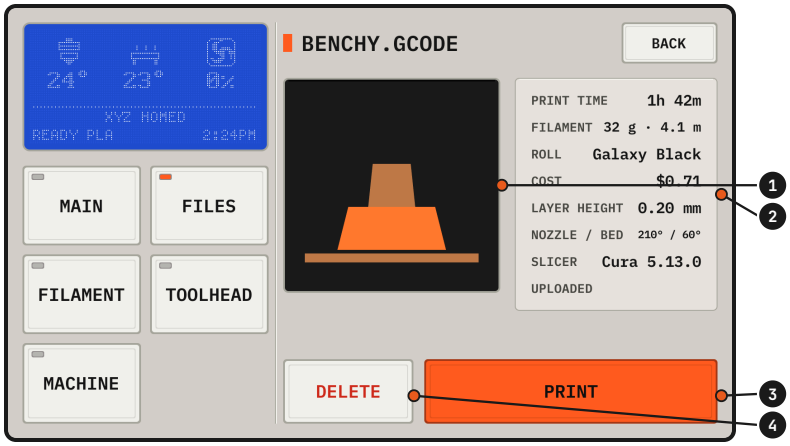
The file list



1. **Count.** The number of files.
2. **Picture.** The thumbnail of the slicer, or the shape of a part.
3. **File.** The name, the print time, the filament and the layer height.
4. **Date.** When the file arrived.

Drag the list up and down. Tap a file to open it.

A file



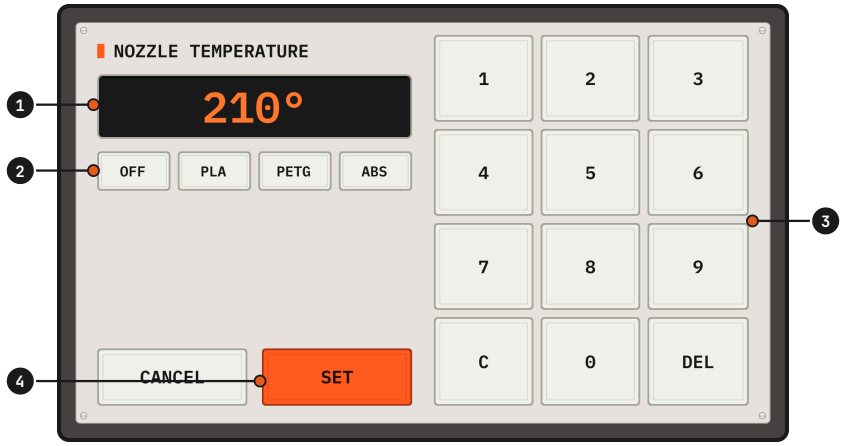
1. **Picture.** The thumbnail of the file.
2. **Slicer data.** The print time, the filament, the layer height, the temperatures, the slicer and the date.
3. **Print.** Starts the print, after a question.
4. **Delete.** Removes the file, after a question. The file that prints now cannot be removed.

Level and calibrate

Maintenance lists the calibrations that your printer has. The list comes from the config of the printer. Open it from Machine.

1. Open Machine, then Maintenance.
2. Tap Bed tilt or Bed mesh to probe the bed. The deck shows the action until it ends.
3. Tap Probe Z offset for the paper test. Move the nozzle down until a sheet of paper drags a little, then tap Accept.
4. Tap PID tune after you change a heater or a fan. A PID tune runs for about 10 minutes.
5. Tap Save config to write the results. Klipper restarts.
6. Tap Pressure advance test to print the tower. Then tap Pressure advance, and type the height of its sharpest corners.

The keypad of a temperature



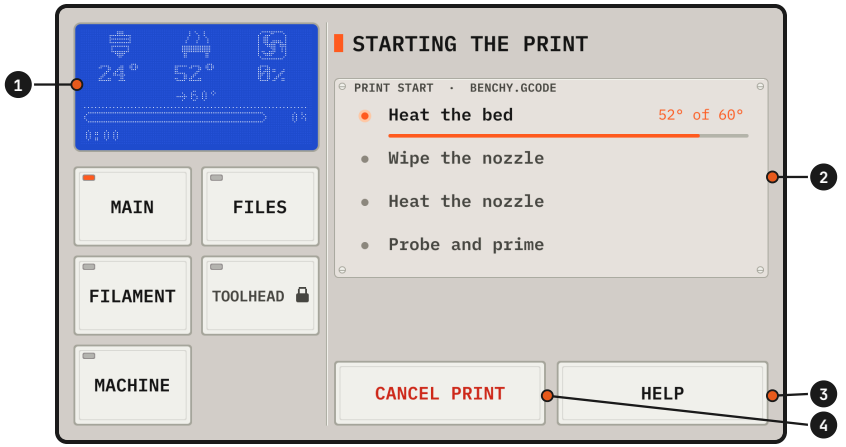
1. **Value.** The temperature that you type.
2. **Presets.** Off, and the temperatures of the materials.
3. **Keys.** The digits. C clears the value, DEL removes the last digit.
4. **Set.** Sets the heater. Cancel closes the keypad with no change.

Watch the print

During a print, the deck shows the screen of the phase of the print. The deck knows the phase from the printer and from the G-code of the file.

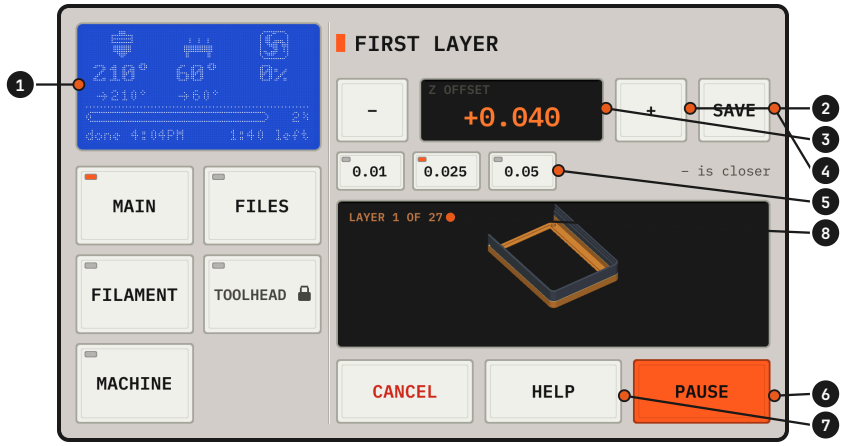
1. Watch the first layer. Use − and + to set the height of the nozzle while it prints.
2. Tap Save to keep the Z offset for the next prints.
3. Use Tune to change the speed, the flow or the fan while the print runs.
4. When the print is done, wait for the bed to cool, then take the part off.

Preparing



1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **Steps.** The steps of the start macro of your printer, in its order. The deck follows the commands that the printer runs: the step that runs has a lit LED, and a done step has a dot.
3. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
4. **Cancel print.** Stops the print, after a question.

The first layer

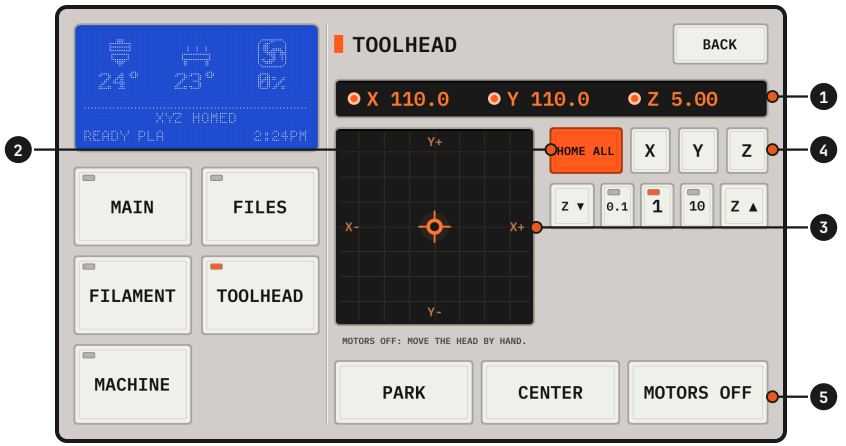


1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **– and +.** – moves the nozzle closer to the bed, + away from it. A line that is too round and thin needs –. A line that is flat and rough needs +.
3. **Z offset.** The height of the nozzle above its normal height. Watch the first lines while they print.
4. **Save.** Writes the Z offset into the config of the printer. Save config in Maintenance then writes the file.
5. **Step.** How far one tap moves the nozzle, in mm.
6. **Pause, Cancel.** Pauses the print, or stops it after a question.
7. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
8. **Layer.** The first layer of the model.

Move the head and heat

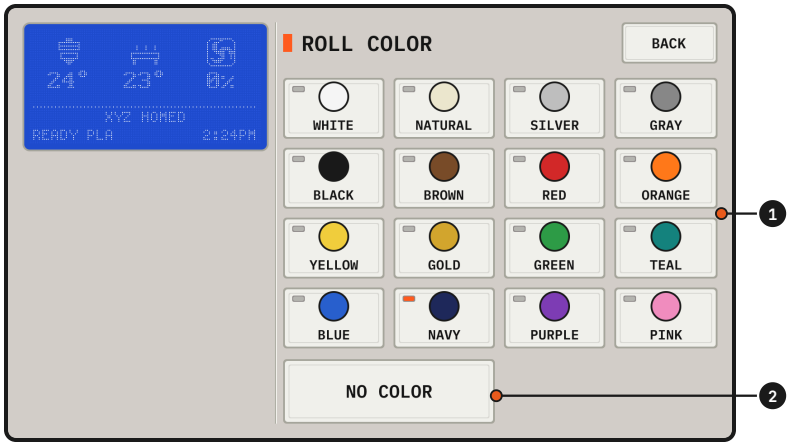
1. Open Toolhead, and tap Home all after each start of the printer.
2. Hold the joystick to move the head. Use Z for the height.
3. Tap a temperature on the idle screen to set it with the keypad.

The Toolhead page



1. **Position.** X, Y and Z of the head. A lit LED shows a homed axis.
2. **Home.** Homes all axes, or one axis. Home after each start of the printer.
3. **Joystick.** Hold a finger on the pad to move the head in X and Y. The farther from the middle, the faster it moves.
4. **Z.** Moves Z up or down by the step: 0.1, 1 or 10 mm.
5. **Park, Center, Motors off.** Park moves the head to the back left. Center moves it to the middle of the bed. Motors off lets you move the head by hand. During a pause, Home, Center and Motors off are off, so that the print can continue. Park lifts the nozzle 10 mm before it moves.

The color of a roll

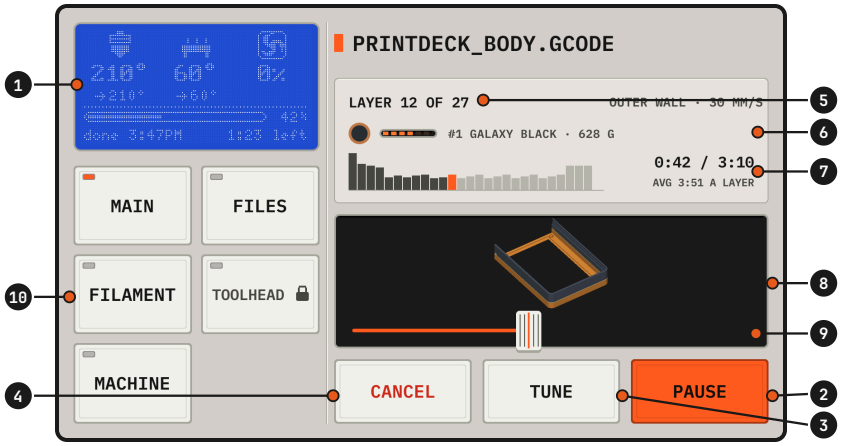


1 **Colors.** The colors of filament. The color of the roll now is lit.

2 **No color.** Removes the color.

A tap on a color sets it, and closes the page. The Cura plugin gives the material of the roll in Cura the same color. A color that you set in Cura comes back to the deck.

Printing



1 **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.

2 **Pause.** Pauses the print.

3 **Tune.** Changes the print while it runs. See Tune.

4 **Cancel.** Stops the print, after a question.

5 **Layer.** The layer that prints now, and the kind of the move now with the live speed of the head.

6 **Roll.** The roll in the printer: a dot in its color, the filament that is left on it, its name, and its grams. The filament of the print counts while it prints. Cura gives a roll its color.

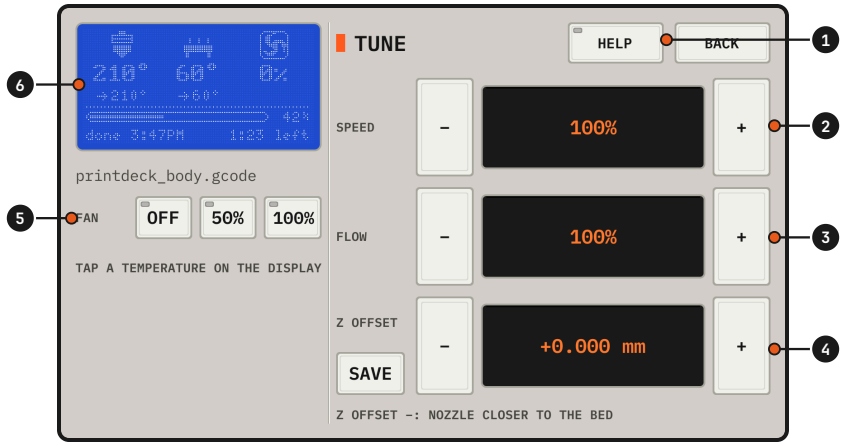
7 **Layer times.** A bar for each layer: its time. The layers that are done are dark, with the time that they took. The layer now is orange, and the layers to come are light, with the time of the slicer. At the right, the time of the layer now and its plan, and the mean time of a layer. A short layer can need more cooling.

8 **Model.** The model up to the layer that prints now, in the color of the roll.

9 **Layer slider.** Drag it to look at another layer. Live goes back to the layer that prints.

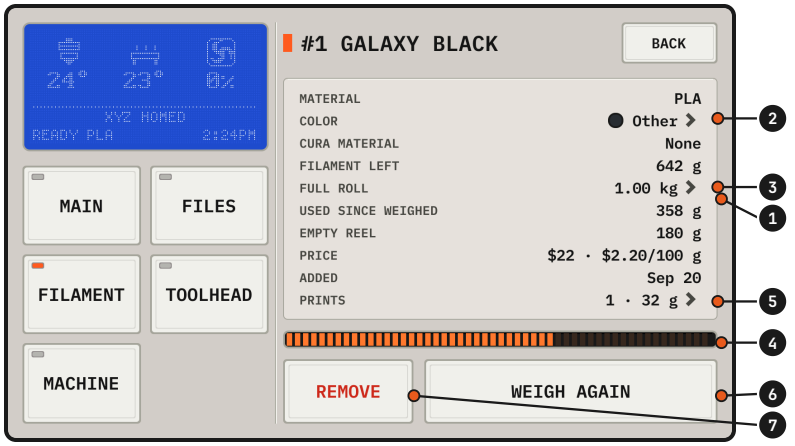
10 **Pages.** During a print, Main is the print. The other keys open their pages while the print goes on, and Main comes back to the print.

Tune



1. **Help.** Opens Wi-Fi and help: the network, the address and the password of the deck.
2. **Speed.** The speed of the print, in percent of the speed of the slicer.
3. **Flow.** The amount of plastic, in percent.
4. **Z offset.** The height of the nozzle above its normal height. Save writes it into the config of the printer, as on the first layer screen.
5. **Fan.** The part cooling fan.
6. **Display.** Tap the nozzle or the bed on the display to change its temperature.

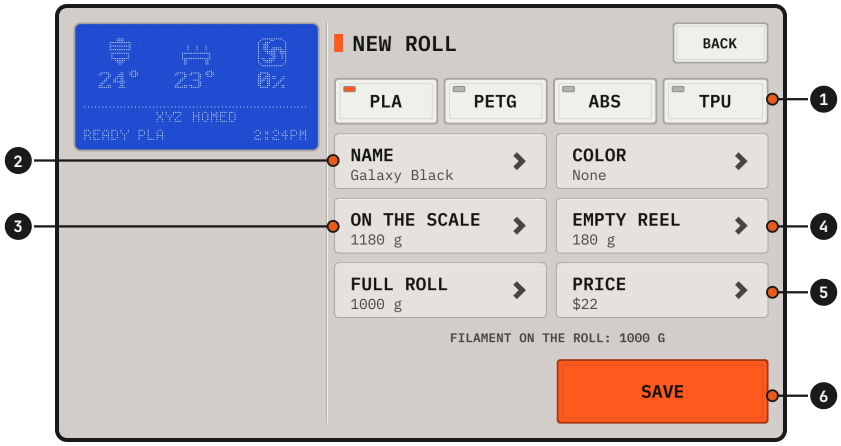
One roll



1. **Details.** The filament on the roll, the filament that prints used, and the price.
2. **Color.** The color of the roll. Tap it to choose a color. The print screen and Cura show it.
3. **Full roll.** The filament of a full roll as it is sold: 1 kg unless you change it. Tap it to type it, for example 250 g or 2 kg. The bar and the price per gram use it, so a roll that you added part used shows how much of a full roll is left.
4. **Left.** The filament on the roll, as a part of a full roll.
5. **Prints.** The number of prints that used the roll, and their filament. Tap it for the list of the prints. A tap on a print opens its page.
6. **Weigh again.** Sets the filament from the weight on a scale.
7. **Remove.** Removes an empty roll from the list.

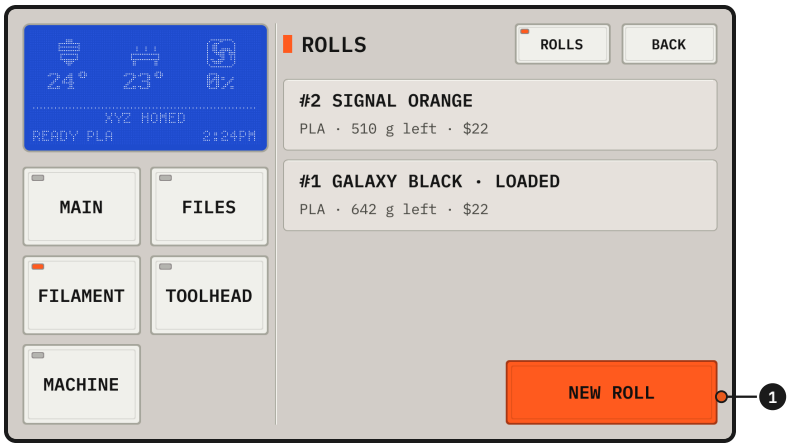
> Each roll has a number, for example #3. The deck does not use a number again. For a label with a QR code, open printdeck.local/rolls/labels in a browser, and print the sheet on round labels of 2 inches (Avery 22807). Choose the first free label of the sheet, so a sheet lasts for many rolls. A phone camera on the label opens the page of the roll: Load opens the filament change on the deck, with the roll.

A new roll



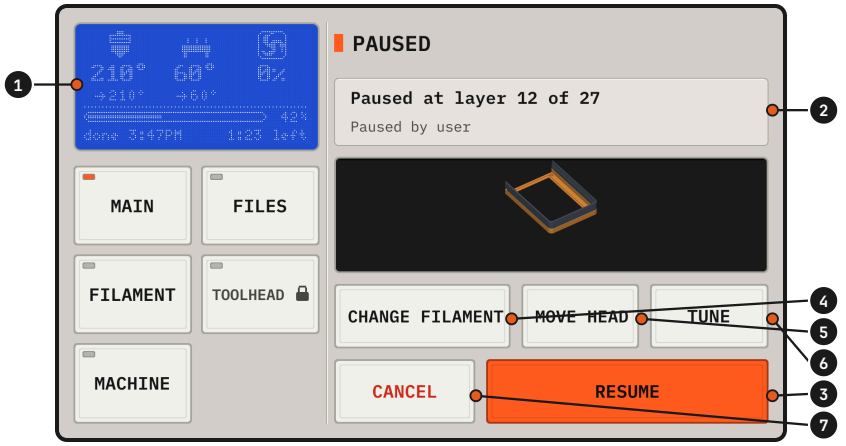
1. **Material.** The material of the roll.
2. **Name.** A name that you type, for example the color and the brand.
3. **On the scale.** The weight of the roll with its reel, in grams.
4. **Empty reel.** The weight of an empty reel of the same brand. The deck keeps the last one.
5. **Price.** The price of the roll in dollars. The deck shows the cost of each print from it.
6. **Save.** Adds the roll and puts it in the printer.

The rolls



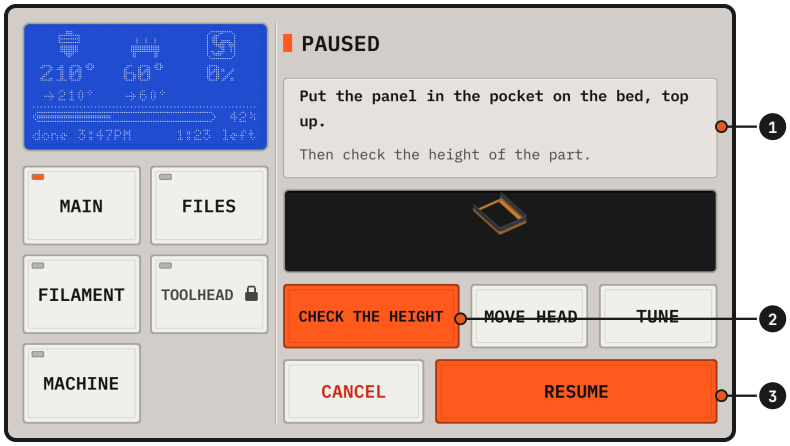
1. **New roll.** Adds a roll without a filament change.

Paused



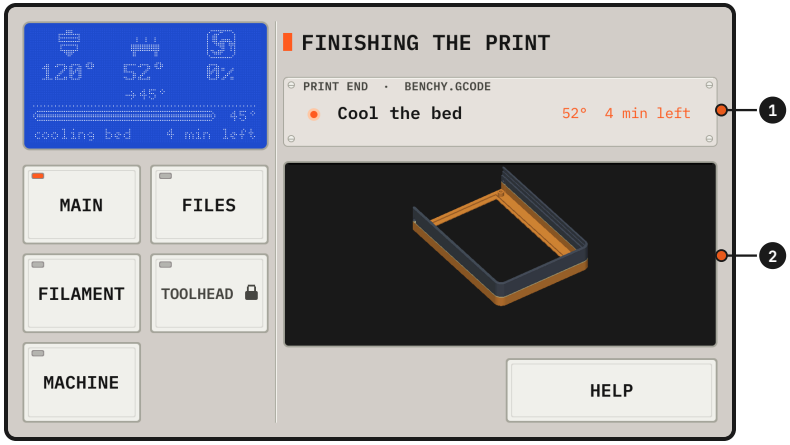
1. **Display.** The status screen of the printer display: the nozzle, the bed and the fan, the progress, the time left, and the time on the clock when the print ends. Tap the nozzle or the bed to set its temperature.
2. **Where.** The layer of the pause, and the message of the printer.
3. **Resume.** Continues the print.
4. **Change filament.** The guided filament change, for a new color. See Change the filament.
5. **Move head.** Opens Toolhead, to move the head out of the way.
6. **Tune.** Changes the print.
7. **Cancel.** Stops the print, after a question.

A pause of the file



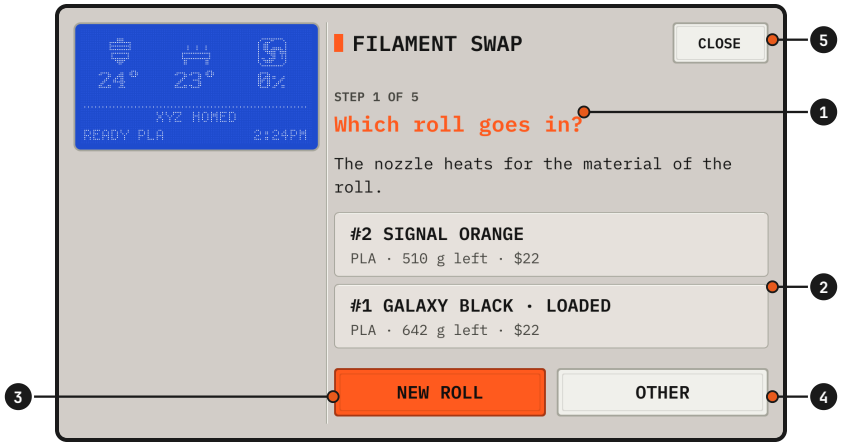
1. **Where.** The message of the file, for example for a part in a pocket on the bed (Print on a part in the Cura plugin).
2. **Check the height.** A paper test on the part. The print goes on from the measured height.
3. **Resume.** Primes the nozzle and continues. With no height check, the deck asks first.

The end of the print



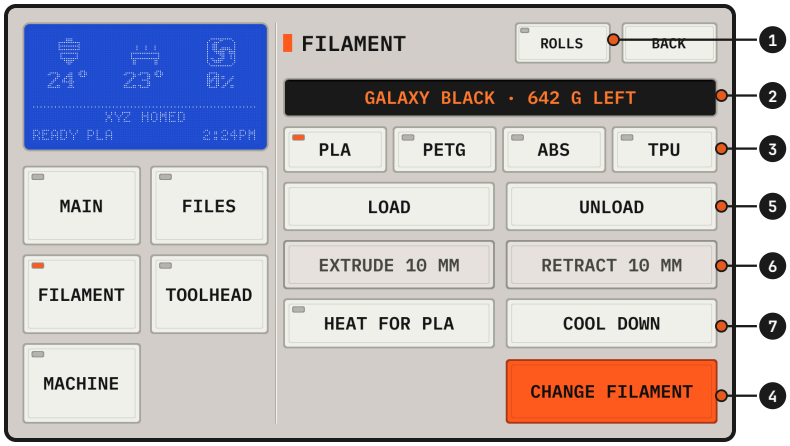
1. **Steps.** The steps of the end macro of your printer, for example the cooling of the bed before the print moves to the front.
2. **Model.** The finished model.

The guided change, step 1



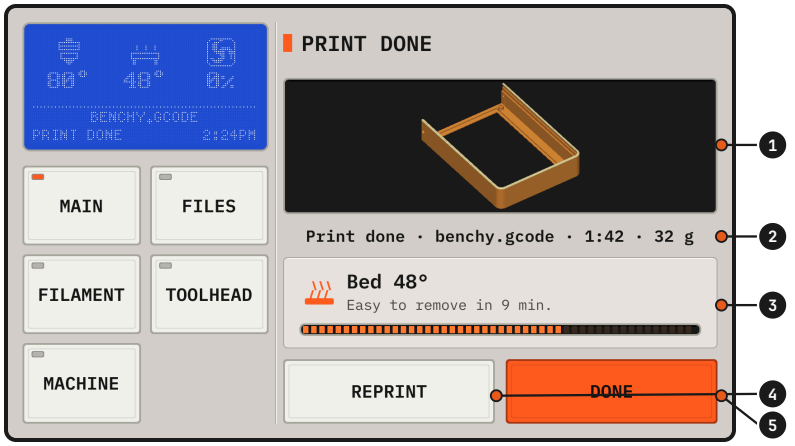
1. **Step.** The step of the change.
2. **Rolls.** Your rolls. Tap the roll that goes in. The nozzle heats for its material.
3. **New roll.** Adds a roll: its material, its name, its weight on a scale, the weight of an empty reel, and its price.
4. **Other.** A filament with no roll: choose the material or type a temperature.
5. **Close.** Stops the change. After the nozzle starts to heat, Close also turns the nozzle off. During a pause, the nozzle goes back to the temperature of the print.

The Filament page



1. **Rolls.** Opens the list of your rolls. On the list, the key stays in its place with its LED lit, and a tap comes back here.
2. **Loaded.** The roll that is in the printer and the filament on it, or the material and its temperatures.
3. **Material.** Tap the material that is in the printer. The deck asks before it changes the loaded material.
4. **Change filament.** The guided change: choose, heat, unload, insert, load, check. The printer homes while the nozzle heats. Turn this off on the Settings page.
5. **Load, Unload.** Only the load, or only the unload.
6. **Extrude, Retract.** Pushes or pulls 10 mm. The nozzle must be at 170° or more.
7. **Heat, Cool down.** Heats the nozzle for the loaded filament, or turns the heaters off.

Done



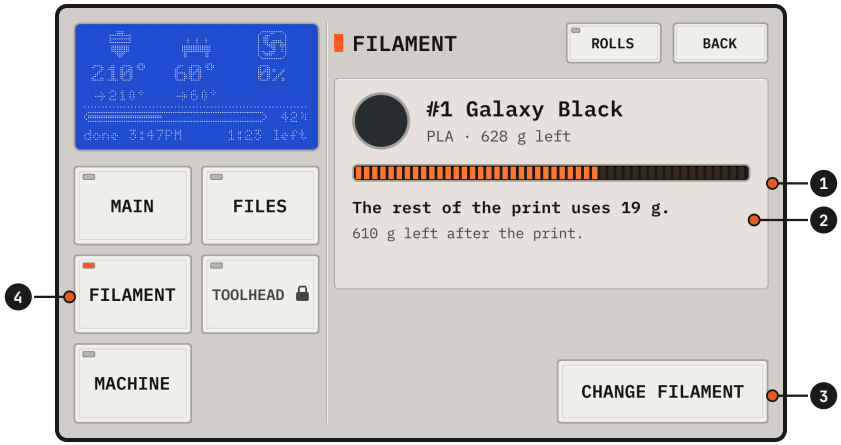
1. **Model.** The model of the print.
2. **Result.** Done, Cancelled or Failed, the file, the time and the filament of the print.
3. **Bed.** The temperature of the bed, and the time until the part comes off easily (35°). The deck learns the time from the cooling after each print. The bar is full at 35°.
4. **Reprint.** Prints the file again.
5. **Done.** Closes this screen.

Look around during a print

The other screens are read only for the printer while it prints: the files, the filament and the rolls, the history. A tap cannot change the print by mistake. The deck goes back to the print at a pause, at a new print, and after 2 minutes with no touch.

During a print, Main is the print. The key panel opens the other pages as at any time, and Main goes back to the print.

Filament during a print



1. **Roll.** The roll in the printer, and the filament that is left on it.
2. **Lasts.** The filament of the rest of the print, and what is left on the roll after it. Red text shows a roll that runs out before the end.
3. **Change filament.** Pauses the print, and opens the filament change. At its end, the nozzle goes back to the temperature of the print: tap Resume.
4. **Pages.** The key panel opens each page, as when the printer does not print.

Change the filament

Set the loaded filament here, so that Preheat uses its temperatures. The guided change heats the nozzle, unloads the old filament and loads the new one.

The deck keeps a list of your rolls of filament. Weigh a new roll with its reel, and the deck subtracts the weight of the empty reel. Each print subtracts the filament that it used from the roll in the printer.

1. Open Filament, then tap Change filament.
2. Tap the roll that goes in, or tap New roll and weigh it. The nozzle heats for its material.
3. Tap Unload, and hold the filament lightly while it comes out.
4. Push the new filament into the extruder until it stops, then tap Load.
5. Check that clean plastic of the new color comes out, then tap Done.